

and attribute similarity into $f(u, v) = A(u, v) + S(u, v)$. Unlike other methods that treat features independently, PrivAGS incorporates clustering features through this composite cohesion measure, with MCEG's cohesion-aware MCMC naturally preserving clustering characteristics through Markov state transitions rather than explicit triangle adjustments. Leveraging community structure as the fundamental granularity for synthesis, combined with BGTm's correlation preservation within communities, PrivAGS exhibits the smallest $RE(C, \tilde{C})$ and RE_{Tri} values across most datasets. While PrivAGS leverages advanced composite cohesion and graph reconstruction techniques to address the high information loss and limited ability of prior methods in capturing tightly connected features, it is still struggle to capture subtle, low-homogeneity topological patterns within the network. Specifically, due to the limitations imposed by privacy budget constraints and the discrete bucket mechanism, coarse-grained community detection and composite cohesion inevitably disrupt delicate fine-grained structural relationships and low-homogeneity topological features. This problem highlight the limitations of current statistical paradigms in modeling complex topological features, underscoring the need for more advanced methods that can comprehensively capture intricate structural patterns in privacy-preserving graph synthesis.

Results in terms of topology structure. Regarding topological structure, PrivAGS shows weaker edge count performance due to distributing its privacy budget across multiple components, causing greater disturbance to global degree distribution. However, it excels in key metrics like the KS statistic and Hellinger distance. PrivAGS enhances topological reconstruction by using communities to extract degree distribution information and recalculating intra- and inter-community distributions. It constructs a seed graph from these distributions to prevent noise injection in composite cohesion distribution. Conversely, AGM-FCL and AGM-Tri reduce global sensitivity from $2n - 2$ to k by truncating node edge counts, yet still introduce substantial noise. Other methods calculate edge probabilities using edge-based attributes while ignoring topological structures, limiting topological expressiveness. AGM-Tri and CPGM-DP's crude triangle adjustments during post-processing impair topological representation, yielding inferior performance versus AGM-FCL.

Results in terms of node attribute. Compared to the baseline method that releases attributes based only on univariate marginal distributions, PrivAGS achieves superior performance (i.e., $NMI(C_P, \tilde{C}_P)$ and $L_1(P, \tilde{P})$) by utilizing multidimensional distributions and a local sampling approach to attribute release. This enhanced performance can be attributed to PrivAGS's innovative approach, which introduces a bounded Gaussian threshold mechanism to more effectively capture intricate dependencies among attributes and construct comprehensive dependency graphs. Moreover, the framework leverages an optimized inference structure as the foundational architecture for distribution inference, which significantly mitigates the impact of noise on the distribution.

7.3 Efficiency Evaluation

Figure 5 presents the average response time results for all methods across all privacy budget and Rényi differential privacy divergence combinations, where PrivAGS^a is configured with hyperparameter $n_b = 500$. Response times are presented as multiples of PrivAGS's time on each dataset. PrivAGS is slightly less efficient than AGM-FCL on the Email dataset due to its heuristic-based edge generation model (MCEG), which prioritizes high composite cohesion edges. These edges concentrate in small regions, causing redundant sampling and reduced efficiency. For other datasets, PrivAGS achieves at least 1.25X efficiency improvement. On large-scale datasets, PrivAGS maintains good synthesis efficiency through density-based compression that enables effective supernode merging, reducing time complexity from computing inter-community edge weight noise. Additionally, MCEG allows efficient edge generation even with low acceptance probabilities on large datasets.